

PHYSICAL SCIENCES

PAPER 1 (PHYSICS)

STUDENT STUDY GUIDE

Everything explained in plain language · Based on past papers



Meet Sol — Your Physics Study Mentor

Sol comes from the Latin word for *sun* — she's all about energy, light, and how the world works. She makes physics feel less like formulas and more like understanding what's happening around you. Calm, logical, quietly confident. Where most students see chaos, Sol sees a formula, a method, and a clear path to full marks. If something doesn't make sense, she breaks it down step by step. No panic — just clean understanding.

WHAT'S INSIDE

- | | |
|----|---|
| P1 | What the Paper Looks Like |
| P2 | The 5 Topics That Matter Most |
| P3 | Step-by-Step Methods with Worked Examples |
| P4 | Mistakes That Cost Students Marks |
| P5 | Your 14-Day Study Plan |

www.rub yaitutor.com



PART 1

What Does the Paper Look Like?

Physical Sciences Paper 1 tests Physics — not Chemistry. It has 10 questions that follow a predictable order every year. Mechanics and circuits dominate. The paper is **not random** — the same topics appear in the same positions. Prepare the right topics and you can walk in knowing exactly what is coming.

Q	What it Tests	Marks	Difficulty
Q1	MCQ: all topics (2 marks each × 10)	20	Easy–Med
Q2	Newton's Laws: FBD, net force, acceleration	16–18	Medium
Q3	Vertical Projectile Motion: v-t graphs, height	14–17	Medium
Q4	Momentum & Impulse: collisions, conservation	12–14	Medium
Q5	Work, Energy & Power: work-energy theorem	12–14	Medium
Q6	Doppler Effect: frequency calculations	11–13	Medium
Q7	Electrostatics: Coulomb's Law, E-fields	13–15	Hard
Q8	Electric Circuits: internal resistance, networks	20–22	Hard
Q9	Electromagnetic Induction / AC generators	13–15	Medium
Q10	Photoelectric Effect: Planck's equation	11–13	Hard

● **Tip:** 150 marks in 3 hours = **1.2 minutes per mark**. Q1 MCQ should take 20 minutes. Q8 circuits gets 25 minutes. If stuck on one part for more than 3 minutes, skip it and come back.



PART 2

The 5 Topics That Give You the Most Marks

These 5 topics appear in **every single paper** and make up over 75 marks out of 150. Master them and you are already halfway there.

1 Newton's Laws & Mechanics

~18
marks

Q2

- Draw a labelled FBD — earns marks on its own
- Apply $F_{\text{net}} = ma$ along EACH axis separately
- Know the difference between static and kinetic friction
- State Newton's laws word-for-word when asked

● **Tip:** ► **Strategy:** FBD + $F_{\text{net}} = ma$. Every time. No exceptions.

2 Electric Circuits & Internal Resistance

~22
marks

Q8

- Parallel: $1/R_p = 1/R_1 + 1/R_2$
- $\text{emf} = I(R + r)$ or $\text{emf} = V_{\text{terminal}} + Ir$
- Terminal voltage drops when current increases — explain in words
- Know how adding/removing resistors changes I and V_T

● **Tip:** ► **Strategy:** Learn the emf equation cold. It appears in almost every circuit question.

3 Work, Energy & Power

~13
marks

Q5

- Work-Energy Theorem: $W_{\text{net}} = \Delta E_k$
- Conservation (no friction): $E_{k1} + E_{p1} = E_{k2} + E_{p2}$
- Power = Work/time = Fv
- Always define your sign convention for direction

● **Tip:** ► **Strategy:** State the Work-Energy Theorem in words — earns 2 marks every time.

4 Momentum & Impulse

~13
marks

Q4

- Conservation: total p before = total p after
- Impulse = $F\Delta t = \Delta p = mv_f - mv_i$
- Define positive direction at the start of every collision question
- Elastic: BOTH momentum AND KE conserved

● **Tip:** ► **Strategy:** Define positive direction first. Then write the conservation equation.

5 Photoelectric Effect

~12
marks

Q10

- $E_{\text{photon}} = hf$ — energy depends on frequency, NOT intensity



- $E_{k(\max)} = hf - W_0$; threshold: $f_0 = W_0/h$
- More intensity = more electrons, same $E_{k(\max)}$
- Below threshold = zero electrons, regardless of brightness

● **Tip: ► Strategy:** $E_{k(\max)}$ depends on frequency only. Intensity changes number, not energy.



PART 3

How to Solve Each Question Type

Follow these steps exactly. They work every time. Each method includes worked examples at three levels — Easy, Standard, and Hard.

METHOD 1

Mechanics & Newton's Laws

Step 1 — Draw the FBD

Draw the object. Add ALL forces as arrows with labels: weight (mg downward), normal force (N , perpendicular to surface), applied force, friction (opposing motion), tension. Each force arrow earns marks on its own.

Step 2 — Choose a positive direction

State clearly: "Take [up / right / down the slope] as positive." This is required by the mark scheme. Never skip it.

Step 3 — Apply $F_{\text{net}} = ma$

Add ALL forces along your chosen axis (with correct signs). Set equal to ma . Solve for the unknown. Remember: on an incline, use $F_{\text{net}} = ma$ along AND perpendicular to the slope.

Step 4 — Check units & direction

Acceleration in $\text{m}\cdot\text{s}^{-2}$. Force in N . If your answer is negative, state the direction clearly: " $2.4 \text{ m}\cdot\text{s}^{-2}$ up the slope."

● **Tip:** Normal force on a horizontal surface: $N = mg$. On an incline: $N = mg \cos\theta$. Always decompose weight first.

WORKED EXAMPLE A

Easy

A 5 kg book rests on a horizontal table. A horizontal force of 30 N is applied. Friction = 8 N. Find the acceleration. ($g = 10 \text{ m}\cdot\text{s}^{-2}$)

Step 1 FBD: Weight = 50 N down, $N = 50 \text{ N}$ up, $F = 30 \text{ N}$ right, $f = 8 \text{ N}$ left.

Step 2 Take right as positive.

Step 3 $F_{\text{net}} = F - f = 30 - 8 = 22 \text{ N}$

Step 4 $F_{\text{net}} = ma \rightarrow 22 = 5 \times a \rightarrow a = 4.4 \text{ m}\cdot\text{s}^{-2}$

✓ **Acceleration = $4.4 \text{ m}\cdot\text{s}^{-2}$ to the right**

WORKED EXAMPLE B

Standard

A 10 kg block is on a rough inclined plane (30°). $\mu_k = 0.2$. Find the acceleration down the slope. $g = 10 \text{ m}\cdot\text{s}^{-2}$

Step 1 Weight component along slope: $mg \sin 30^\circ = 10 \times 10 \times 0.5 = 50 \text{ N}$ (down slope)

Step 2 Normal: $N = mg \cos 30^\circ = 10 \times 10 \times 0.866 = 86.6 \text{ N}$

Step 3 Friction: $f = \mu_k N = 0.2 \times 86.6 = 17.32 \text{ N}$ (up slope, opposing motion)

Step 4 $F_{\text{net}} = 50 - 17.32 = 32.68 \text{ N}$ down slope



Step 5 $a = F_{\text{net}}/m = 32.68/10 = 3.27 \text{ m}\cdot\text{s}^{-2}$ down the slope

✓ **Acceleration = $3.27 \text{ m}\cdot\text{s}^{-2}$ down the slope**

WORKED EXAMPLE C

Hard

Two blocks: A (8 kg) on a rough horizontal surface ($\mu_k = 0.3$), connected by a light string over a frictionless pulley to B (5 kg) hanging. $g = 10 \text{ m}\cdot\text{s}^{-2}$. Find: (a) acceleration, (b) tension.

Step 1 For B (taking down as positive): $5g - T = 5a \rightarrow 50 - T = 5a \dots (1)$

Step 2 Normal on A: $N = 8g = 80 \text{ N}$. Friction on A: $f = 0.3 \times 80 = 24 \text{ N}$

Step 3 For A (taking right as positive): $T - f = 8a \rightarrow T - 24 = 8a \dots (2)$

Step 4 Add (1) and (2): $50 - 24 = 13a \rightarrow a = 2.0 \text{ m}\cdot\text{s}^{-2}$

Step 5 Substitute into (1): $T = 50 - 5(2) = 50 - 10 = 40 \text{ N}$

✓ **Acceleration = $2.0 \text{ m}\cdot\text{s}^{-2}$; Tension = 40 N**

Try It Yourself

A 3 kg block rests on a horizontal surface. A force of 20 N is applied at 30° above horizontal. $\mu_k = 0.2$. Find the acceleration. ($g = 10$)

Answer: $F_x = 20\cos 30^\circ = 17.32 \text{ N}$; $N = 30 - 20\sin 30^\circ = 20 \text{ N}$; $f = 0.2 \times 20 = 4 \text{ N}$;
 $F_{\text{net}} = 17.32 - 4 = 13.32 \text{ N}$; $a = 13.32/3 = 4.44 \text{ m}\cdot\text{s}^{-2}$



**METHOD
2****Momentum & Collisions****Step 1 — Define positive direction**

State it explicitly: "Let right (east) be positive." Any velocity to the left gets a negative sign.

Step 2 — Write the conservation equation

$$p_{\text{total before}} = p_{\text{total after}} \rightarrow m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

Step 3 — Substitute with signs

Replace each variable with its value and correct sign. If objects stick together: $v_{1f} = v_{2f} = v_f$.

Step 4 — Check elastic/inelastic

Elastic: KE before = KE after (rare in NSC). Inelastic: KE not conserved. Calculate both and compare to check.

● **Tip:** Impulse = $F\Delta t = \Delta p = mv_f - mv_i$. This formula connects force, time and momentum change.

WORKED EXAMPLE A**Easy**

A 1200 kg car travels east at $20 \text{ m}\cdot\text{s}^{-1}$ and collides with a stationary 800 kg car. They move together. Find the velocity after collision.

Step 1 Let east = positive.

Step 2 Conservation: $(1200)(20) + (800)(0) = (1200+800)v_f$

Step 3 $24\,000 = 2000 v_f$

Step 4 $v_f = 12 \text{ m}\cdot\text{s}^{-1}$ east

✓ $v_f = 12 \text{ m}\cdot\text{s}^{-1}$ east

WORKED EXAMPLE B**Standard**

Car A (1200 kg , $+20 \text{ m}\cdot\text{s}^{-1}$) hits Car B (800 kg , $-15 \text{ m}\cdot\text{s}^{-1}$). After: A moves at $+5 \text{ m}\cdot\text{s}^{-1}$. Find B's final velocity.

Step 1 Let east = positive.

Step 2 $(1200)(20) + (800)(-15) = (1200)(5) + (800)v_{Bf}$

Step 3 $24\,000 - 12\,000 = 6\,000 + 800v_{Bf}$

Step 4 $12\,000 - 6\,000 = 800v_{Bf} \rightarrow v_{Bf} = 7.5 \text{ m}\cdot\text{s}^{-1}$ east

✓ $v_{Bf} = 7.5 \text{ m}\cdot\text{s}^{-1}$ east

WORKED EXAMPLE C**Hard**

A 0.15 kg cricket ball is hit at $30 \text{ m}\cdot\text{s}^{-1}$ left. It rebounds at $40 \text{ m}\cdot\text{s}^{-1}$ right. Contact time = 0.005 s . Find: (a) impulse, (b) force on bat.

Step 1 Let right = positive. $v_i = -30 \text{ m}\cdot\text{s}^{-1}$; $v_f = +40 \text{ m}\cdot\text{s}^{-1}$

Step 2 Impulse = $m(v_f - v_i) = 0.15(40 - (-30)) = 0.15(70)$



Step 3 Impulse = 10.5 N·s to the right (on the ball)

Step 4 $F = \text{Impulse}/\Delta t = 10.5/0.005 = 2100 \text{ N}$ on ball (right)

Step 5 Newton's 3rd law: Force on bat = 2100 N to the LEFT

✓ **Impulse = 10.5 N·s; Force on bat = 2100 N**

 Try It Yourself

A 60 kg skater (+8 m·s⁻¹) collides with a stationary 40 kg skater. They move together. Find v_f .

Answer: $(60)(8) + (40)(0) = (100)v_f \rightarrow v_f = 4.8 \text{ m·s}^{-1}$



**METHOD
3****Work, Energy & Power****Step 1 — State the theorem**

Work-Energy Theorem: $W_{\text{net}} = \Delta E_k = E_{k_f} - E_{k_i}$. Writing this earns marks every time.

Step 2 — List all work done

$W_{\text{net}} = W_{\text{applied}} + W_{\text{friction}} + W_{\text{gravity}}$. Friction work is ALWAYS negative (opposes motion).
Gravity work = mgh (positive going down, negative going up).

Step 3 — Apply conservation if no friction

$E_{\text{mechanical}} = E_k + E_p = \text{constant}$. $\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$

Step 4 — Calculate power

$P = W/t = Fv$. Use average velocity if speed changes. Check units: watts (W) or kilowatts (kW).

● **Tip:** $W = Fs \cos\theta$ — only the component of force PARALLEL to displacement does work. If force $\perp$ displacement, $W = 0$.

WORKED EXAMPLE A**Easy**

A 5 kg box slides 4 m across a floor. Friction = 10 N. Find work done by friction.

Step 1 $W = Fs \cos\theta$. Friction opposes motion, so $\theta = 180^\circ$.

Step 2 $W_{\text{friction}} = 10 \times 4 \times \cos 180^\circ = 10 \times 4 \times (-1)$

Step 3 $W_{\text{friction}} = -40 \text{ J}$

✓ **$W_{\text{friction}} = -40 \text{ J}$**

WORKED EXAMPLE B**Standard**

A 60 kg skier starts from rest at the top of an 80 m slope. At the bottom, $v = 35 \text{ m}\cdot\text{s}^{-1}$. $g = 10 \text{ m}\cdot\text{s}^{-2}$. Find work done by friction.

Step 1 $W_{\text{net}} = \Delta E_k = \frac{1}{2}mv_f^2 - 0 = \frac{1}{2}(60)(35)^2 = 36\,750 \text{ J}$

Step 2 $W_{\text{gravity}} = mgh = 60 \times 10 \times 80 = 48\,000 \text{ J}$

Step 3 $W_{\text{net}} = W_{\text{gravity}} + W_{\text{friction}}$

Step 4 $36\,750 = 48\,000 + W_{\text{friction}}$

Step 5 $W_{\text{friction}} = -11\,250 \text{ J}$

✓ **$W_{\text{friction}} = -11\,250 \text{ J}$ (negative = energy lost to heat)**

WORKED EXAMPLE C**Hard**

A 1200 kg car engine exerts 4000 N. Friction = 800 N. Car starts from rest and travels 500 m. Find: (a) KE gained, (b) power at $20 \text{ m}\cdot\text{s}^{-1}$.

Step 1 $W_{\text{net}} = (F_{\text{engine}} - f) \times d = (4000 - 800) \times 500 = 1\,600\,000 \text{ J}$

Step 2 $\Delta E_k = W_{\text{net}} = 1\,600\,000 \text{ J}$

Step 3 Check: $v^2 = 2as \rightarrow a = 3200/1200 = 2.67 \text{ m}\cdot\text{s}^{-2}$; $v^2 = 2(2.67)(500) = 2667$; $E_k = \frac{1}{2}(1200)(2667) \approx 1\,600\,000 \text{ J}$ ✓



Step 4 $P = F \times v = 4000 \times 20 = 80\,000\text{ W} = 80\text{ kW}$

✓ **KE = 1 600 000 J; Power = 80 kW**

 Try It Yourself

A 2 kg ball falls 5 m from rest (no friction). $g = 10\text{ m}\cdot\text{s}^{-2}$. Find its speed at the bottom using energy conservation.

Answer: $mgh = \frac{1}{2}mv^2 \rightarrow (2)(10)(5) = \frac{1}{2}(2)v^2 \rightarrow v^2 = 100 \rightarrow v = 10\text{ m}\cdot\text{s}^{-1}$



**METHOD
4****Electric Circuits & Internal Resistance****Step 1 — Find total external resistance**

Series: $R_{\text{total}} = R_1 + R_2 + \dots$ | Parallel: $1/R_p = 1/R_1 + 1/R_2 + \dots$ In mixed circuits: find parallel combination first, then add in series.

Step 2 — Apply emf equation

emf (ϵ) = $I(R_{\text{ext}} + r)$ or $\epsilon = V_{\text{terminal}} + Ir$ (where r = internal resistance). Solve for the unknown current I .

Step 3 — Find voltage across each component

$V = IR$ for each resistor. For parallel branches: voltage is the SAME across each branch. For series elements: current is the SAME through each.

Step 4 — Power & terminal voltage

$P = VI = I^2R = V^2/R$. Terminal voltage $V_T = \epsilon - Ir$. V_T decreases when I increases (more current = more internal voltage drop).

⚠ Warning: When external resistance **DECREASES**, current **INCREASES**, and terminal voltage **DECREASES**. This is the most commonly tested concept in Q8 — know it in words **AND** with numbers.

WORKED EXAMPLE A**Easy**

Battery: emf = 12 V, $r = 0.5 \Omega$, external $R = 5.5 \Omega$. Find: (a) current, (b) terminal voltage.

Step 1 $I = \epsilon / (R + r) = 12 / (5.5 + 0.5) = 12 / 6 = 2 \text{ A}$

Step 2 $V_{\text{terminal}} = \epsilon - Ir = 12 - (2)(0.5) = 12 - 1 = 11 \text{ V}$

Step 3 Check: $V_T = IR_{\text{ext}} = 2 \times 5.5 = 11 \text{ V} \checkmark$

✓ I = 2 A; $V_{\text{terminal}} = 11 \text{ V}$

WORKED EXAMPLE B**Standard**

emf = 9 V, $r = 1 \Omega$. $R_1 = 3 \Omega$ in series with parallel combination of $R_2 = 4 \Omega$ and $R_3 = 6 \Omega$. Find: current through R_2 .

Step 1 Parallel: $1/R_p = 1/4 + 1/6 = 5/12 \rightarrow R_p = 2.4 \Omega$

Step 2 $R_{\text{ext}} = 3 + 2.4 = 5.4 \Omega$

Step 3 $I_{\text{total}} = 9 / (5.4 + 1) = 9 / 6.4 = 1.41 \text{ A}$

Step 4 $V_{\text{parallel}} = I_{\text{total}} \times R_p = 1.41 \times 2.4 = 3.38 \text{ V}$

Step 5 $I_{R_2} = V_{\text{parallel}} / R_2 = 3.38 / 4 = 0.84 \text{ A}$

✓ I through $R_2 = 0.84 \text{ A}$

WORKED EXAMPLE C**Hard**

A battery with internal resistance r is connected to $R = 4 \Omega$. Ammeter reads 2 A. Terminal voltage = 8 V. Find: (a) emf, (b) internal resistance r , (c) power dissipated internally.

Step 1 Check: $V_T = IR_{\text{ext}} = 2 \times 4 = 8 \text{ V} \checkmark$ (terminal voltage confirmed)



Step 2 $\epsilon = V_T + Ir \rightarrow$ also: $V_{\text{internal}} = \epsilon - 8$

Step 3 From $V = IR_{\text{ext}}$: $I = 2 \text{ A}$ confirmed. We need another reading or emf given.

Step 4 If emf = 10 V: $r = (\epsilon - V_T)/I = (10-8)/2 = 1 \Omega$

Step 5 $P_{\text{internal}} = I^2 r = (2)^2(1) = 4 \text{ W}$

✓ **emf = 10 V; $r = 1 \Omega$; $P_{\text{internal}} = 4 \text{ W}$**

Try It Yourself

A 6 V battery ($r = 0.5 \Omega$) is connected to two resistors of 4Ω and 6Ω in parallel. Find the total current drawn from the battery.

Answer: $1/R_p = 1/4 + 1/6 = 5/12 \rightarrow R_p = 2.4 \Omega$; $I = 6/(2.4+0.5) = 6/2.9 = 2.07 \text{ A}$



DOPPLER EFFECT**Step 1 — Identify motion**

Is the SOURCE moving toward or away from the observer? Toward observer: observed frequency INCREASES. Away from observer: observed frequency DECREASES.

Step 2 — Apply the formula

$f_L = f_s \times (v \pm v_L) / (v \mp v_s)$. v = speed of sound ($340 \text{ m}\cdot\text{s}^{-1}$). Top: + when listener moves toward source. Bottom: – when source moves toward listener.

PHOTOELECTRIC EFFECT**Step 3 — State the key equation**

$E_{\text{photon}} = hf = hc/\lambda$ and $E_{k(\text{max})} = hf - W_0$ (where W_0 = work function). $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$. Threshold: $f_0 = W_0/h$.

Step 4 — Analyse intensity vs frequency

More intensity = more photons = more electrons ejected. Higher frequency = more energy per photon = higher $E_{k(\text{max})}$. Below threshold frequency: NO electrons ejected, regardless of intensity.

● **Tip:** NSC loves asking: 'Light intensity doubles. What happens to $E_{k(\text{max})}$?' Answer: $E_{k(\text{max})}$ stays the SAME. Only frequency changes it.

WORKED EXAMPLE A

Easy

An ambulance siren emits 1200 Hz. It moves toward you at $30 \text{ m}\cdot\text{s}^{-1}$. Speed of sound = $340 \text{ m}\cdot\text{s}^{-1}$. Find the frequency you hear.

Step 1 Source moving toward observer: use – in denominator.

Step 2 $f_L = 1200 \times 340 / (340 - 30) = 1200 \times 340 / 310$

Step 3 $f_L = 1200 \times 1.0968 = 1316 \text{ Hz}$

✓ $f_L \approx 1316 \text{ Hz}$ (higher than emitted)

WORKED EXAMPLE B

Standard

Light of frequency $8.0 \times 10^{14} \text{ Hz}$ hits a metal surface. Work function $W_0 = 6.0 \times 10^{-19} \text{ J}$. $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$. Find $E_{k(\text{max})}$ of emitted electrons.

Step 1 $E_{\text{photon}} = hf = (6.63 \times 10^{-34})(8.0 \times 10^{14}) = 5.304 \times 10^{-19} \text{ J}$

Step 2 $E_{k(\text{max})} = hf - W_0 = 5.304 \times 10^{-19} - 6.0 \times 10^{-19}$

Step 3 Wait — $E_{\text{photon}} < W_0$: NO electrons emitted (frequency below threshold)

Step 4 Check threshold: $f_0 = W_0/h = 6.0 \times 10^{-19} / 6.63 \times 10^{-34} = 9.05 \times 10^{14} \text{ Hz}$

Step 5 $8.0 \times 10^{14} \text{ Hz} < 9.05 \times 10^{14} \text{ Hz} \rightarrow$ below threshold $\rightarrow$ no emission

✓ No photoelectrons emitted — frequency is below threshold



WORKED EXAMPLE C**Hard**

Zinc has threshold wavelength 290 nm. Light of wavelength 200 nm shines on it. Find $E_{k(\max)}$ in eV. ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$, $c = 3 \times 10^8 \text{ m}\cdot\text{s}^{-1}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$)

Step 1 $W_0 = hc/\lambda_0 = (6.63 \times 10^{-34})(3 \times 10^8)/(290 \times 10^{-9}) = 6.86 \times 10^{-19} \text{ J}$

Step 2 $E_{\text{photon}} = hc/\lambda = (6.63 \times 10^{-34})(3 \times 10^8)/(200 \times 10^{-9}) = 9.945 \times 10^{-19} \text{ J}$

Step 3 $E_{k(\max)} = 9.945 \times 10^{-19} - 6.86 \times 10^{-19} = 3.085 \times 10^{-19} \text{ J}$

Step 4 In eV: $E_{k(\max)} = 3.085 \times 10^{-19} / 1.6 \times 10^{-19} = 1.93 \text{ eV}$

✓ $E_{k(\max)} = 3.09 \times 10^{-19} \text{ J} = 1.93 \text{ eV}$

 Try It Yourself

A source emits sound at 800 Hz and moves AWAY from you at $20 \text{ m}\cdot\text{s}^{-1}$. Speed of sound = $340 \text{ m}\cdot\text{s}^{-1}$. Find the frequency you hear.

Answer: $f_L = 800 \times 340/(340+20) = 800 \times 340/360 = 755.6 \text{ Hz}$



PART 4

Mistakes That Cost Students Marks

These are the most common reasons marks are dropped — even by students who understand the work.

1. Not Stating Direction or Sign Convention

✗ Writing " $a = 3.5 \text{ m}\cdot\text{s}^{-2}$ " without a direction, or using momentum values without defining positive direction.

✓ State your positive direction at the TOP of every mechanics and momentum answer. Write: "Let right be positive." Every vector answer needs a direction.

2. Using the Wrong Formula for the Situation

✗ Using $v = u + at$ (no friction, constant acceleration) when friction is present, or using energy conservation when there IS non-conservative work.

✓ Decide first: Is there friction or external forces? If yes, use the Work-Energy Theorem. If no friction: conservation of mechanical energy. Never mix the two.

3. Missing Units in the Final Answer

✗ Writing " $F = 24$ " instead of " $F = 24 \text{ N}$ " loses a mark in every calculation question.

✓ Write units after EVERY number in your working, not just the final answer. Units: force = N, energy = J, power = W, momentum = $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$.

4. Calculator in Wrong Mode (Degrees vs Radians)

✗ Doppler and wave questions sometimes involve angles. If your calculator is set to radians, trigonometry answers will be completely wrong.

✓ Before any trig calculation: press MODE → confirm DEG. This single check prevents a cascade of wrong answers.

5. Forgetting Internal Resistance in Circuit Problems

✗ Setting $I = V/R_{\text{ext}}$ and ignoring the internal resistance r . The correct formula is $I = \text{emf}/(R_{\text{ext}} + r)$.

✓ When you see the word "battery" and "internal resistance r " in any circuit question, use $I = \text{emf}/(R_{\text{ext}} + r)$. Always. Not optional.

6. Confusing Intensity and Frequency in Photoelectric Questions

✗ Writing "increasing intensity increases $E_{k(\text{max})}$ " — this is wrong and costs 2 marks.

✓ Intensity = more photons = more electrons. Frequency = more energy per photon = higher $E_{k(\text{max})}$. These are two completely separate effects. Memorise the difference.



PART 5

Your 14-Day Study Plan

Follow this plan day by day. Topics earlier in the plan are worth the most marks. Do not skip ahead — each block builds on the previous one.

Days 1-3

Newton's Laws & Mechanics

~18
marks

- Open any past Paper 1. Find Q2.
- Draw 15 free body diagrams from different scenarios.
- Apply $F_{\text{net}} = ma$ to 20 different questions (horizontal, inclined, connected).
- Practise questions involving friction on slopes.
- Goal: Set up any mechanics question in under 3 minutes.

Days 4-5

Momentum & Impulse

~13
marks

- Write the conservation of momentum equation 10 times until it is automatic.
- Practise 15 collision questions — always define positive direction first.
- Practise 10 impulse questions: find force, time, or change in momentum.
- Test yourself: elastic vs inelastic — calculate KE before and after.
- Goal: Any collision question solved in 5 minutes with correct direction.

Days 6-7

Work, Energy & Power

~13
marks

- Write the Work-Energy Theorem from memory. Know it word-for-word.
- Practise 10 problems where friction is present (use $W_{\text{net}} = \Delta E_k$).
- Practise 10 conservation problems where no friction ($E_{\text{mechanical}} = \text{constant}$).
- Practise calculating power: $P = W/t$ and $P = Fv$.
- Goal: Full marks on Q5 every time — it is consistent and formulaic.

Days 8-10

Electric Circuits & Internal Resistance

~22
marks

- Master the emf equation: $\epsilon = I(R + r)$. Write it 20 times.
- Practise 5 series-parallel combination questions per day.
- For each circuit: find I_{total} , then V across each component, then P .
- Practise the "what happens if R_3 is removed" type analysis questions.
- Goal: A complete Q8-style question solved accurately in 20 minutes.

**Days
11-12**

Doppler & Photoelectric Effect

~25
marks
combined

- Practise Doppler: source toward = frequency up; away = frequency down.



- Practise 10 Doppler calculations using the formula both directions.
- Photoelectric: memorise that intensity $\neq E_k$. Write it on a card.
- Practise 10 $E_{k(\max)} = hf - W_0$ calculations.
- Goal: Full marks on Q6 and Q10 — these are the most formulaic questions.

Day 13	Timed Mixed Practice	<i>Test yourself</i>
---------------	-----------------------------	----------------------

- Set a timer. Do Q1 MCQ (20 min). Q2 (18 min). Q5 (15 min). Q8 (25 min).
- Work in silence. Do not check notes.
- After: mark your answers. Write exactly where you lost marks.
- Redo only the parts you got wrong — full working, not just the answer.

Day 14	Full Past Paper — Exam Simulation	<i>Exam conditions</i>
---------------	--	------------------------

- Print a full past Paper 1 (most recent year available).
- Set 3-hour timer. Silence your phone. No notes.
- When done: mark it fully. Write your total score.
- List 3 specific things to fix. Revise those before the real exam.

EVERY DAY

Daily 5-Minute Warm-Up

Do this before every study session to keep all skills sharp.

Daily Task	Why
✓ 3 $F_{\text{net}} = ma$ calculations	Keeps Newton's Laws automatic — no thinking needed in the exam.
✓ 1 momentum conservation problem	Builds speed. Define direction, write equation, solve.
✓ Write the emf equation from memory	Prevents the most costly circuit mistake in the paper.
✓ 1 Work-Energy Theorem question	Keeps the substitution pattern fresh and accurate.
✓ 1 Photoelectric rule: intensity vs frequency	Say it out loud — hearing yourself helps you remember.



You can do this. The paper is predictable. The methods are learnable. Master the 5 core topics, avoid the 6 common mistakes, and follow the 14-day plan. That is all you need to do.

www.rub yaitutor.com · Physical Sciences P1 Study Pack



**PREDICTIVE 2026 PHYSICAL SCIENCES PAPER 1
(PHYSICS)****PAPER A — QUESTION PAPER****Time:** 3 hours**Total Marks:** 150**Date:** 2026**READ THE FOLLOWING INSTRUCTIONS CAREFULLY:**

1. This question paper consists of 10 questions. Answer ALL questions.
2. Start each question on a new page.
3. Number the answers correctly according to the numbering system in this paper.
4. Leave one line between sub-questions.
5. Show ALL formulae and substitutions in ALL calculations.
6. Round off your answers to TWO decimal places where applicable.
7. Give brief motivations, discussions, et cetera where required.
8. You may use a non-programmable calculator.
9. Data and formulae sheet is provided on the last page.

QUESTION 1*Multiple Choice — Choose ONE answer for each question.*

1.1 A ball is thrown vertically upward. At its highest point, the net force on the ball is: **(2)**

A zero B downward C upward D equal to the weight

1.2 Two forces of 3 N and 4 N act perpendicularly on an object. The resultant force is: **(2)**

A 1 N B 5 N C 7 N D 12 N

1.3 The change in momentum of an object equals: **(2)**

A the net force on it B the impulse on it C the work done on it D its kinetic energy

1.4 An object moving with velocity v has kinetic energy E . If its velocity doubles, its kinetic energy becomes: **(2)**

A $E/2$ B $2E$ C $4E$ D $\sqrt{2}E$

1.5 A source of sound moves toward a stationary observer. The frequency heard by the observer compared to emitted frequency is: **(2)**

A higher B lower C the same D zero

1.6 Two point charges Q and $2Q$ are separated by a distance r . The electrostatic force between them is F . If the distance is doubled, the force becomes: **(2)**

A $F/4$ B $F/2$ C $2F$ D $4F$

1.7 In an electric circuit, a voltmeter should be connected: **(2)**

A in series with the component B in parallel with the component C in series with the battery D between the battery terminals only

1.8 The rms voltage of an AC supply is 220 V. The peak voltage is approximately: **(2)**

A 155 V B 220 V C 311 V D 440 V

1.9 The photoelectric effect is best explained by treating light as: **(2)**

A a wave B a particle (photon) C neither wave nor particle D an electron stream

1.10 A photon has energy E . If the frequency of the photon is doubled, its energy becomes:

(2)

A $E/2$ B E C $2E$ D $4E$

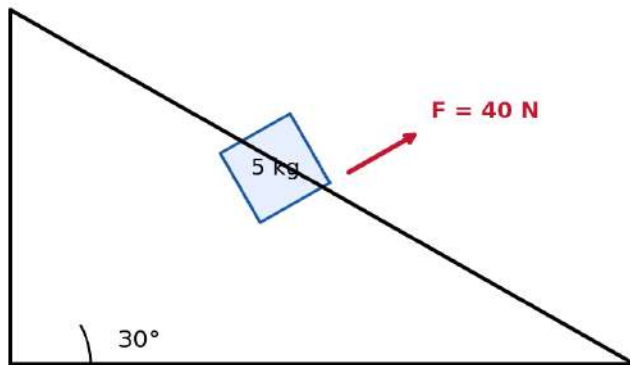
[20]

QUESTION 2

Newton's Laws and Forces

A 5 kg block is placed on a rough inclined plane making an angle of 30° with the horizontal. A force of 40 N is applied parallel to and up the incline. The coefficient of kinetic friction is 0.2. Take $g = 10 \text{ m}\cdot\text{s}^{-2}$.

Q2: A 5 kg block on a 30° inclined plane.
A force of 40 N is applied up the slope. ($\mu_k = 0.2$)



- 2.1 Draw a labelled free body diagram (FBD) for the block showing all forces acting on it. (4)
- 2.2 Calculate the normal force N acting on the block. (3)
- 2.3 Calculate the magnitude of the frictional force acting on the block. (2)
- 2.4 Calculate the acceleration of the block. (4)
- 2.5 State Newton's Second Law of Motion in words. (2)
- 2.6 A second identical block is placed on top of the first. How does this affect the acceleration of the system? Explain your answer. (1)

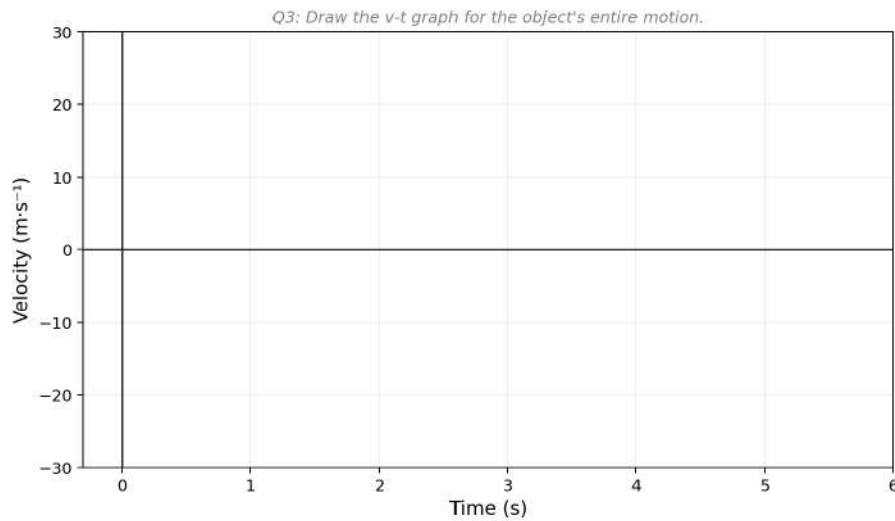
[16]

QUESTION 3

Vertical Projectile Motion

A ball is thrown vertically upward from the ground with an initial velocity of $25 \text{ m}\cdot\text{s}^{-1}$. Ignore air resistance. Take upward as positive. $g = 10 \text{ m}\cdot\text{s}^{-2}$ (downward).

- 3.1 Calculate the maximum height reached by the ball. (3)
- 3.2 Calculate the time taken to reach maximum height. (2)
- 3.3 Calculate the total time of flight (from launch until it returns to the ground). (2)
- 3.4 Draw a velocity-time (v - t) graph for the full motion of the ball. Use the axes provided below. Label all key values. (4)



3.5 At what time(s) is the speed of the ball equal to $15 \text{ m}\cdot\text{s}^{-1}$? Show all calculations. **(4)**

3.6 A second ball is dropped from rest from the same maximum height at the instant the first ball begins its downward journey. Calculate the height above the ground when the two balls meet. **(2)**

[17]

QUESTION 4

Momentum and Impulse

A 1 200 kg car travelling east at $20 \text{ m}\cdot\text{s}^{-1}$ collides with a 800 kg car travelling west at $15 \text{ m}\cdot\text{s}^{-1}$. After the collision, the 1 200 kg car continues east at $5 \text{ m}\cdot\text{s}^{-1}$.

4.1 Calculate the velocity of the 800 kg car after the collision. State the direction of your answer. **(5)**

4.2 Is this collision elastic or inelastic? Show calculations to support your answer. **(4)**

4.3 Calculate the impulse experienced by the 800 kg car during the collision. **(3)**

[12]

QUESTION 5

Work, Energy and Power

A 60 kg skier starts from rest at the top of a ski slope 80 m high. At the bottom of the slope her speed is $35 \text{ m}\cdot\text{s}^{-1}$. Take $g = 10 \text{ m}\cdot\text{s}^{-2}$.

5.1 Calculate the gravitational potential energy of the skier at the top of the slope. **(2)**

5.2 Calculate the kinetic energy of the skier at the bottom of the slope. **(2)**

5.3 Calculate the magnitude of the work done by friction on the skier. **(3)**

5.4 State the work-energy theorem. **(2)**

5.5 The skier travels 200 m along the slope from top to bottom. Calculate the average frictional force acting on the skier. **(2)**

5.6 The skier reaches the bottom of the slope in 8 seconds. Calculate the average power output of the motor tow-lift that would bring her back to the top (same 200 m, taking 40 s). Ignore friction on the way up. **(2)**

[13]

QUESTION 6

Doppler Effect

An ambulance siren emits a sound with a frequency of 1 200 Hz. The speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$. A stationary observer stands by the roadside.

- 6.1** State the Doppler effect. (2)
- 6.2** The ambulance approaches the observer at $30 \text{ m}\cdot\text{s}^{-1}$. Calculate the frequency heard by the observer. (3)
- 6.3** The ambulance passes the observer and moves away at the same speed. Calculate the frequency now heard by the observer. (3)
- 6.4** How would a graph of observed frequency vs. time look as the ambulance passes the observer? Sketch and describe. (2)
- 6.5** Give ONE application of the Doppler effect in medicine. (2)

[12]

QUESTION 7

Electrostatics

Three point charges are placed in a line: charge $Q_1 = +4 \times 10^{-6} \text{ C}$ at point A, charge $Q_2 = -6 \times 10^{-6} \text{ C}$ at point B (0.3 m from A), and point C is 0.2 m to the right of B. $k = 9 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$.

- 7.1** State Coulomb's Law. (2)
- 7.2** Calculate the magnitude of the electrostatic force between Q_1 and Q_2 . (3)
- 7.3** A small positive test charge $Q_3 = +1 \times 10^{-6} \text{ C}$ is placed at C. Calculate the net force on Q_3 . State the direction. (5)
- 7.4** Draw the electric field pattern between Q_1 and Q_2 . Show direction of field lines. (4)

[14]

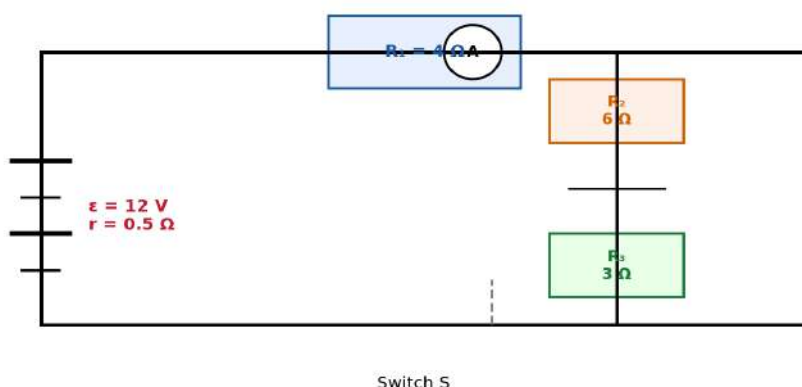
QUESTION 8

Electric Circuits

A battery with emf 12 V and internal resistance $0.5\ \Omega$ is connected to an external circuit. $R_1 = 4\ \Omega$ is connected in series, and $R_2 = 6\ \Omega$ and $R_3 = 3\ \Omega$ are connected in parallel with each other and in series with R_1 . Refer to the circuit diagram.

Paper A — Q8: Electric Circuit (Internal Resistance)

$$\mathcal{E} = 12\text{ V}, r = 0.5\ \Omega$$



- 8.1 Calculate the equivalent resistance of R_2 and R_3 in parallel. (2)
- 8.2 Calculate the total resistance in the external circuit. (2)
- 8.3 Calculate the current delivered by the battery. (3)
- 8.4 Calculate the terminal voltage of the battery. (3)
- 8.5 Calculate the potential difference across the parallel combination ($R_2 \parallel R_3$). (3)
- 8.6 Calculate the current through R_2 . (3)
- 8.7 Calculate the power dissipated in R_1 . (2)
- 8.8 How does the brightness of R_1 change if R_3 is disconnected? Explain. (4)

[22]

QUESTION 9

Electromagnetic Induction and AC

An AC generator produces an alternating EMF described by $e = 200 \sin(100\pi t)$ volts, where t is in seconds.

- 9.1 State Faraday's Law of electromagnetic induction. (2)
- 9.2 Write down the peak (maximum) EMF. (1)
- 9.3 Calculate the rms EMF of the generator. (2)
- 9.4 Calculate the frequency of the AC. (2)
- 9.5 A resistor of $50\ \Omega$ is connected to this generator. Calculate the rms current through the resistor. (2)
- 9.6 Calculate the average power dissipated by the resistor. (2)

9.7 Compare an AC generator and a DC generator. State ONE difference.

(2)

[13]

QUESTION 10

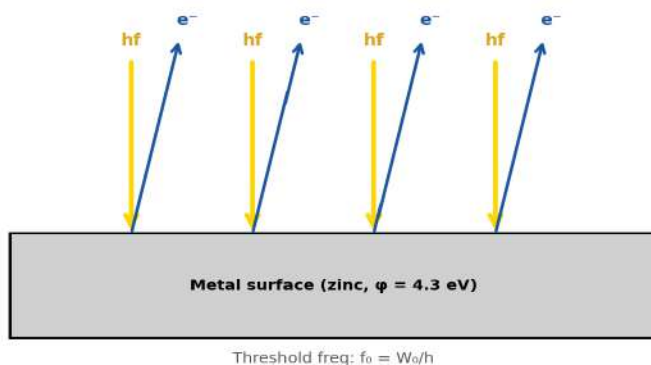
Photoelectric Effect

Light of frequency 8.0×10^{14} Hz shines on a zinc metal surface. The work function of zinc is 6.88×10^{-19} J. Planck's constant $h = 6.63 \times 10^{-34}$ J·s.

Paper A — Q10: Photoelectric Effect

$$E_k(\text{max}) = hf - \phi = hf - W_0$$

W_0 = work function = 4.3 eV = 6.88×10^{-19} J



- 10.1 Define the term "work function" of a metal. (2)
- 10.2 Calculate the energy of ONE photon of the incident light. (2)
- 10.3 Calculate the maximum kinetic energy of the emitted photoelectrons. (3)
- 10.4 Calculate the threshold frequency of zinc. (2)
- 10.5 If the intensity of the light is doubled but the frequency remains the same, explain the effect on: (a) the number of photoelectrons emitted, and (b) the maximum kinetic energy of each photoelectron. (2)

[11]

TOTAL: 150 marks

PREDICTIVE 2026 PHYSICAL SCIENCES PAPER 1 (PHYSICS)

PAPER A — MEMORANDUM

Accept alternative correct methods. Penalise 1 mark per formula not stated. Award marks for substitution even if final answer is incorrect (consequential marks).

QUESTION 1 [20 marks]

- 1.1 B — downward (weight acts down; at highest point $v=0$ but g still acts) (2)
- 1.2 B — 5 N (Pythagoras: $\sqrt{3^2+4^2} = \sqrt{25} = 5$ N) (2)
- 1.3 B — the impulse on it (Impulse = $F\Delta t = \Delta p$) (2)
- 1.4 C — 4E ($E \propto v^2$; if v doubles, $E \times 2^2 = 4E$) (2)
- 1.5 A — higher (Doppler effect: moving source toward observer = higher frequency) (2)
- 1.6 A — $F/4$ ($F \propto 1/r^2$; if r doubles, $F \times 1/4$) (2)
- 1.7 B — in parallel with the component (2)
- 1.8 C — 311 V ($V_{\text{peak}} = V_{\text{rms}} \times \sqrt{2} = 220 \times 1.414 \approx 311$ V) (2)
- 1.9 B — a particle (photon) (2)
- 1.10 C — 2E ($E = hf$; if f doubles, E doubles) (2)

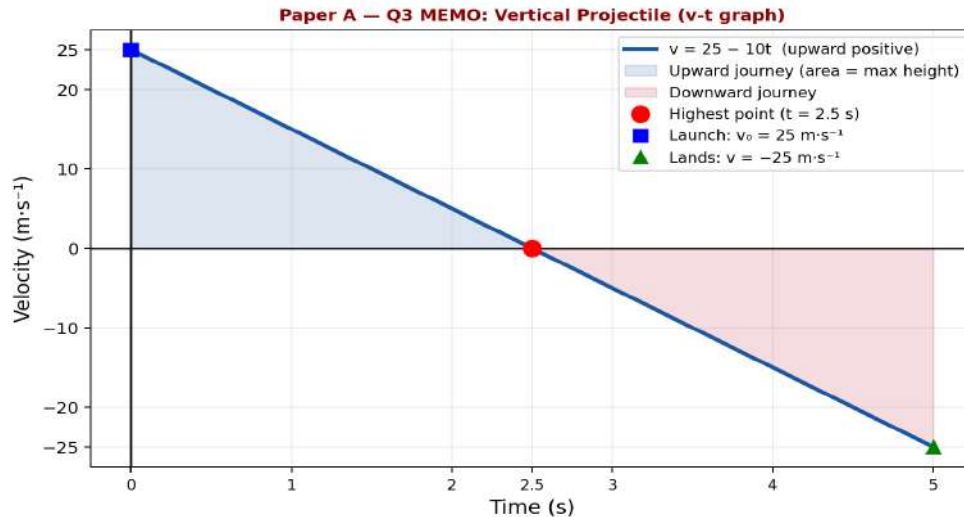
[TOTAL Q1: 20]

QUESTION 2 [16 marks]

- 2.1 FBD must show: Weight W downward ($W = mg = 50$ N) (4)
Normal force N perpendicular to slope (upward from slope)
Applied force $F = 40$ N up the slope
Friction force f down the slope (kinetic, opposing motion)
- 2.2 Components perpendicular to incline: $N - W \cos 30^\circ = 0$ (3)
 $N = mg \cos 30^\circ = 5 \times 10 \times \cos 30^\circ = 50 \times 0.866$
 $N = 43.30$ N
- 2.3 $f_k = \mu_k \times N = 0.2 \times 43.30$ (2)
 $f_k = 8.66$ N (directed down the slope)
- 2.4 Net force up slope = $F - f_k - mg \sin 30^\circ$ (4)
 $= 40 - 8.66 - 50 \times 0.5 = 40 - 8.66 - 25$
 $= 6.34$ N (up the slope)
 $a = F_{\text{net}}/m = 6.34/5$
 $a = 1.27 \text{ m}\cdot\text{s}^{-2}$ up the slope
- 2.5 Newton's Second Law: The net (resultant) force acting on an object is equal to the product of the mass of the object and its acceleration. The acceleration is in the same direction as the net force. ($F_{\text{net}} = ma$) (2)
- 2.6 N doubles, so friction doubles; $mg \sin 30^\circ$ doubles; F unchanged. (1)
Total mass doubles so net force per unit mass stays the same
→ acceleration unchanged.

[TOTAL Q2: 16]

QUESTION 3 [17 marks]



- 3.1** $v^2 = v_0^2 + 2a\Delta y$; at max height $v = 0$ (3)
 $0 = (25)^2 + 2(-10)(\Delta y)$
 $\Delta y = 625/20$
 Maximum height = 31.25 m
- 3.2** $v = v_0 + at$; $v = 0$ (2)
 $0 = 25 - 10t$
 $t = 2.5$ s
- 3.3** Total time = $2 \times$ time to top = 2×2.5 (2)
 Total flight time = 5 s
- 3.4** v-t graph: straight line starting at (0; +25), crossing zero at $t = 2.5$ s, ending at (5; -25) (4)
 ✓ correct shape ✓ axes labelled ✓ key values (0, 25, 2.5, 5, -25)
- 3.5** Speed = 15 m·s⁻¹ means $v = +15$ or $v = -15$ (4)
 Case 1: $15 = 25 - 10t \rightarrow t = 1.0$ s
 Case 2: $-15 = 25 - 10t \rightarrow t = 4.0$ s
 Speed equals 15 m·s⁻¹ at $t = 1.0$ s (going up) and $t = 4.0$ s (coming down)
- 3.6** Second ball drops from rest at $h = 31.25$ m when $t = 2.5$ s (2)
 Let τ = time after second ball released
 Ball 1 (coming down from max): $y_1 = 31.25 - 5\tau^2$
 Ball 2 (falling): $y_2 = 31.25 - 5\tau^2$
 Wait — both have same expression: $y = 31.25 - 5\tau^2$
 They meet immediately! Both balls are at max height at the same instant.
 Meeting height = 31.25 m (they meet at the highest point)

[TOTAL Q3: 17]

QUESTION 4 [12 marks]

- 4.1 Take east as positive. (5)
 $p_i = m_1 v_{1i} + m_2 v_{2i} = 1200(+20) + 800(-15) = 24000 - 12000 = 12000 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$
 $p_f = m_1 v_{1f} + m_2 v_{2f} = 1200(+5) + 800v_{2f} = 6000 + 800v_{2f}$
Conservation of momentum: $12000 = 6000 + 800v_{2f}$
 $v_{2f} = 6000/800 = 7.5 \text{ m}\cdot\text{s}^{-1}$
The 800 kg car moves EAST at $7.5 \text{ m}\cdot\text{s}^{-1}$ after the collision
- 4.2 $E_{ki} = \frac{1}{2}(1200)(20^2) + \frac{1}{2}(800)(15^2) = 240000 + 90000 = 330000 \text{ J}$ (4)
 $E_{kf} = \frac{1}{2}(1200)(5^2) + \frac{1}{2}(800)(7.5^2) = 15000 + 22500 = 37500 \text{ J}$
Since $E_{kf} < E_{ki}$ (kinetic energy is NOT conserved): INELASTIC collision
- 4.3 Impulse on 800 kg car = $\Delta p = m(v_f - v_i) = 800(7.5 - (-15))$ (3)
 $= 800 \times 22.5$
Impulse = 18000 N·s eastward
- [TOTAL Q4: 12]

QUESTION 5 [13 marks]

- 5.1 $EP = mgh = 60 \times 10 \times 80$ (2)
 $EP = 48000 \text{ J} = 48 \text{ kJ}$
- 5.2 $EK = \frac{1}{2}mv^2 = \frac{1}{2} \times 60 \times 35^2 = \frac{1}{2} \times 60 \times 1225$ (2)
 $EK = 36750 \text{ J} = 36.75 \text{ kJ}$
- 5.3 By energy conservation: $EP(\text{top}) = EK(\text{bottom}) + W_{\text{friction}}$ (3)
(lost to heat)
 $48000 = 36750 + |W_{\text{friction}}|$
 $W_{\text{friction}} = 11250 \text{ J}$ (work done against friction)
- 5.4 The work-energy theorem: The net work done on an object (2)
equals the change in the kinetic energy of the object. ($W_{\text{net}} = \Delta EK = EK_f - EK_i$)
- 5.5 $W_{\text{friction}} = f \times d$ (friction force acts opposite to displacement) (2)
 $11250 = f \times 200$
 $f = 56.25 \text{ N}$
- 5.6 Work done lifting skier = $mgh = 48000 \text{ J}$ (2)
 $P = W/t = 48000/40$
 $P = 1200 \text{ W} = 1.2 \text{ kW}$
- [TOTAL Q5: 13]

QUESTION 6 [12 marks]

- 6.1 The Doppler effect is the change in frequency (or pitch) of a wave experienced by an observer when the source of the wave and the observer are in relative motion. (2)
- 6.2 $f_L = v/(v - v_s) \times f_s$ (source approaching) (3)
 $= 340/(340 - 30) \times 1200$
 $= 340/310 \times 1200$
 $f_L = 1316 \text{ Hz}$ (approximately)
- 6.3 $f_L = v/(v + v_s) \times f_s$ (source moving away) (3)
 $= 340/(340 + 30) \times 1200$
 $= 340/370 \times 1200$
 $f_L = 1103 \text{ Hz}$ (approximately)
- 6.4 The graph shows a step-down: frequency starts high ($\approx 1316 \text{ Hz}$), then drops suddenly to a lower value ($\approx 1103 \text{ Hz}$) as the ambulance passes. ✓ high frequency section ✓ drop at passing point ✓ lower section after (2)
- 6.5 Sonar/ultrasound to measure blood flow velocity (echocardiography). OR: measuring foetal heartbeat using Doppler ultrasound. (2)

[TOTAL Q6: 12]

QUESTION 7 [14 marks]

- 7.1 Coulomb's Law: The magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. $F = kQ_1Q_2/r^2$ (2)
- 7.2 $F = kQ_1Q_2/r^2 = (9 \times 10^9 \times 4 \times 10^{-6} \times 6 \times 10^{-6}) / (0.3)^2$ (3)
 $= (9 \times 10^9 \times 24 \times 10^{-12}) / 0.09$
 $= 216 \times 10^{-3} / 0.09$
 $F = 2.4 \text{ N}$
- 7.3 C is 0.2 m from B (Q_2) and 0.5 m from A (Q_1). (5)
Force from Q_1 on Q_3 : $F_1 = k(4 \times 10^{-6})(1 \times 10^{-6})/(0.5)^2 = 9 \times 10^9 \times 4 \times 10^{-12} / 0.25 = 0.144 \text{ N}$ (repulsive: rightward)
Force from Q_2 on Q_3 : $F_2 = k(6 \times 10^{-6})(1 \times 10^{-6})/(0.2)^2 = 9 \times 10^9 \times 6 \times 10^{-12} / 0.04 = 1.35 \text{ N}$ (attractive: leftward)
Net force = $F_2 - F_1 = 1.35 - 0.144 = 1.206 \text{ N}$ leftward (toward Q_2)
- 7.4 Electric field lines: start at positive Q_1 , curve toward negative Q_2 (since Q_2 is negative) (4)
Lines point from Q_1 to Q_2 . More dense near larger charge (Q_2).
✓ lines from + to - ✓ correct density ✓ no lines crossing

[TOTAL Q7: 14]

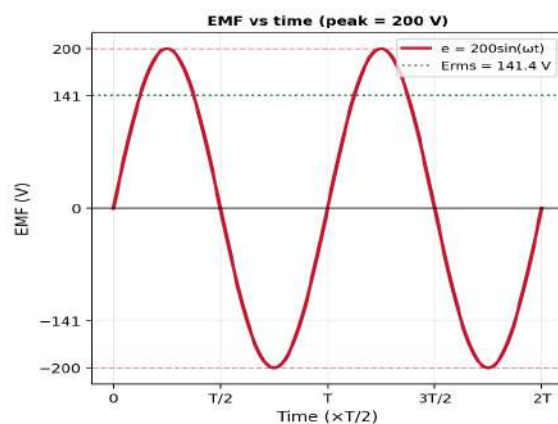
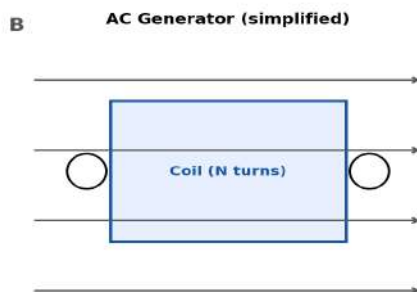
QUESTION 8 [22 marks]

- 8.1 $1/R_{\text{parallel}} = 1/R_2 + 1/R_3 = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$ (2)
 $R_{\text{parallel}} = 2 \Omega$
- 8.2 $R_{\text{external}} = R_1 + R_{\text{parallel}} = 4 + 2 = 6 \Omega$ (2)
- 8.3 $I = \varepsilon / (R_{\text{ext}} + r) = 12 / (6 + 0.5) = 12 / 6.5$ (3)
 $I = 1.85 \text{ A}$
- 8.4 $V_{\text{terminal}} = \varepsilon - Ir = 12 - 1.85 \times 0.5 = 12 - 0.923$ (3)
 $V_{\text{terminal}} = 11.08 \text{ V}$
- 8.5 $V_{\text{parallel}} = I \times R_{\text{parallel}} = 1.85 \times 2$ (3)
 $V_{\text{parallel}} = 3.69 \text{ V}$
- 8.6 $I_2 = V_{\text{parallel}} / R_2 = 3.69 / 6$ (3)
 $I_2 = 0.62 \text{ A}$
- 8.7 $P_1 = I^2 R_1 = (1.85)^2 \times 4 = 3.4225 \times 4$ (2)
 $P_1 = 13.69 \text{ W}$
- 8.8 Disconnecting R_3 increases R_{parallel} (from 2Ω to 6Ω) → (4)
 R_{total} increases (from 6Ω to 10.5Ω)
 $\therefore$ Total current I decreases → voltage across $R_1 = IR_1$ decreases
 Therefore R_1 becomes DIMMER (less power dissipated in R_1)

[TOTAL Q8: 22]

QUESTION 9 [13 marks]

Paper A — Q9 MEMO: AC Generator & Induced EMF



- 9.1 Faraday's Law: The magnitude of the induced EMF across the (2)
 ends of a conductor is proportional to the rate of change of
 the magnetic flux linkage through the conductor.
- 9.2 Peak EMF = 200 V (the amplitude of $e = 200 \sin(100\pi t)$) (1)
- 9.3 $E_{\text{rms}} = E_{\text{peak}} / \sqrt{2} = 200 / \sqrt{2} = 200 / 1.414$ (2)
 $E_{\text{rms}} = 141.42 \text{ V}$
- 9.4 $\omega = 100\pi \text{ rad}\cdot\text{s}^{-1}$, and $\omega = 2\pi f$ (2)
 $f = \omega / 2\pi = 100\pi / 2\pi$
 $f = 50 \text{ Hz}$
- 9.5 $I_{\text{rms}} = E_{\text{rms}} / R = 141.42 / 50$ (2)
 $I_{\text{rms}} = 2.83 \text{ A}$
- 9.6 $P = I_{\text{rms}} \times E_{\text{rms}} = 2.83 \times 141.42 = 400 \text{ W}$ (or $P = E_{\text{rms}}^2 / R =$ (2)
 $20000 / 50 = 400 \text{ W}$)

- 9.7 AC generator uses slip rings → current alternates direction (AC output). (2)
DC generator uses a split-ring commutator → current always flows in one direction (DC output).

[TOTAL Q9: 13]

QUESTION 10 [11 marks]

- 10.1 The work function (W_0) of a metal is the minimum energy required to remove an electron from the surface of that metal. (It is the binding energy of the most loosely bound electron.) (2)
- 10.2 $E = hf = 6.63 \times 10^{-34} \times 8.0 \times 10^{14}$ (2)
 $E = 5.304 \times 10^{-19} \text{ J}$
- 10.3 $E_k(\text{max}) = E - W_0 = 5.304 \times 10^{-19} - 6.88 \times 10^{-19}$ (3)
 $= -1.576 \times 10^{-19} \text{ J (NEGATIVE!)}$
Since $E_k(\text{max}) < 0$, no photoelectrons are emitted.
The frequency of incident light is BELOW the threshold frequency.
- 10.4 $f_0 = W_0/h = 6.88 \times 10^{-19} / 6.63 \times 10^{-34}$ (2)
 $f_0 = 1.038 \times 10^{15} \text{ Hz}$
- 10.5 (a) Number of photoelectrons: INCREASES (more photons per second hit the surface) (2)
(b) Maximum kinetic energy: UNCHANGED ($E_k = hf - W_0$ depends on frequency, not intensity)

[TOTAL Q10: 11]

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	TOTAL
20	16	17	12	13	12	14	22	13	11	150

PREDICTIVE 2026 PHYSICAL SCIENCES PAPER 1 (PHYSICS)

PAPER B — QUESTION PAPER

Time: 3 hours

Total Marks: 150

Date: 2026

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 10 questions. Answer ALL questions.
2. Start each question on a new page.
3. Show ALL formulae and substitutions in ALL calculations.
4. Round off to TWO decimal places where applicable.
5. Give brief motivations/discussions where required.
6. You may use a non-programmable calculator.
7. Data and formulae sheet provided on the last page.

QUESTION 1

Multiple Choice — Choose ONE answer for each question.

1.1 A book lies on a table. The reaction force to the weight of the book (Newton's 3rd law pair) is: (2)

A The normal force from the table B The gravitational force the book exerts on Earth C The weight of the table D The normal force on the table

1.2 An object slides down a frictionless inclined plane. The net force on it equals: (2)

A mg B $mg \sin \theta$ C $mg \cos \theta$ D zero

1.3 A ball is thrown upward and caught at the same height. During the flight, which quantity remains constant? (2)

A speed B velocity C acceleration D kinetic energy

1.4 The law of conservation of momentum states that in a collision: (2)

A kinetic energy is always conserved B total momentum is conserved if no external net force acts C total momentum always increases D momentum is transferred to heat

1.5 A car decelerates uniformly from $30 \text{ m} \cdot \text{s}^{-1}$ to rest over 150 m. The deceleration is: (2)

A $1 \text{ m} \cdot \text{s}^{-2}$ B $2 \text{ m} \cdot \text{s}^{-2}$ C $3 \text{ m} \cdot \text{s}^{-2}$ D $5 \text{ m} \cdot \text{s}^{-2}$

1.6 An ambulance moves away from a stationary person. The person hears the siren at a frequency that is: (2)

A higher than emitted B lower than emitted C the same as emitted D zero

1.7 The electric field strength at a distance r from a point charge Q is: (2)

A kQ/r^2 B kQ^2/r C kQ/r D kQ^2/r^2

1.8 In a circuit, the terminal voltage of a battery equals the emf when: (2)

A the external resistance is very small B no current flows C the internal resistance is very large D the circuit is short-circuited

1.9 Which phenomenon provides evidence that light has a particle nature? (2)

A Diffraction B Interference C Photoelectric effect D Polarisation

1.10 The threshold frequency of a metal is f_0 . If light of frequency $2f_0$ shines on the surface, the maximum kinetic energy of emitted electrons is: (2)

A hf_0 B $2hf_0$ C $h(2f_0 - f_0) = hf_0$ D zero

[20]

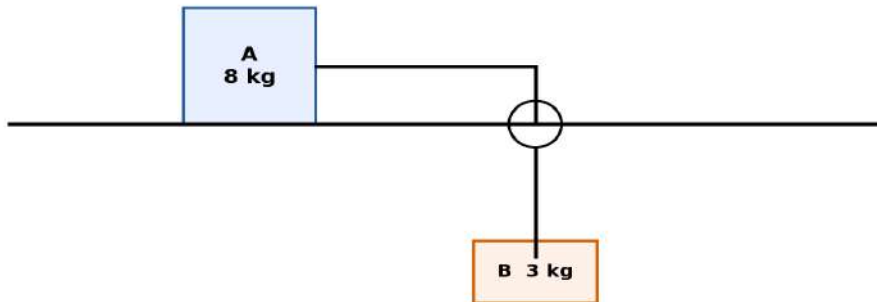
QUESTION 2

Newton's Laws and Forces

Block A (8 kg) rests on a rough horizontal surface and is connected by a light inextensible string over a frictionless pulley to Block B (3 kg) which hangs freely. The coefficient of kinetic friction between A and the surface is 0.25. Take $g = 10 \text{ m}\cdot\text{s}^{-2}$. The system is released from rest.

Q2: Block A (8 kg) on a horizontal surface is connected by a light string over a frictionless pulley to Block B (3 kg) hanging freely.

$\mu_k = 0.25$ between A and surface



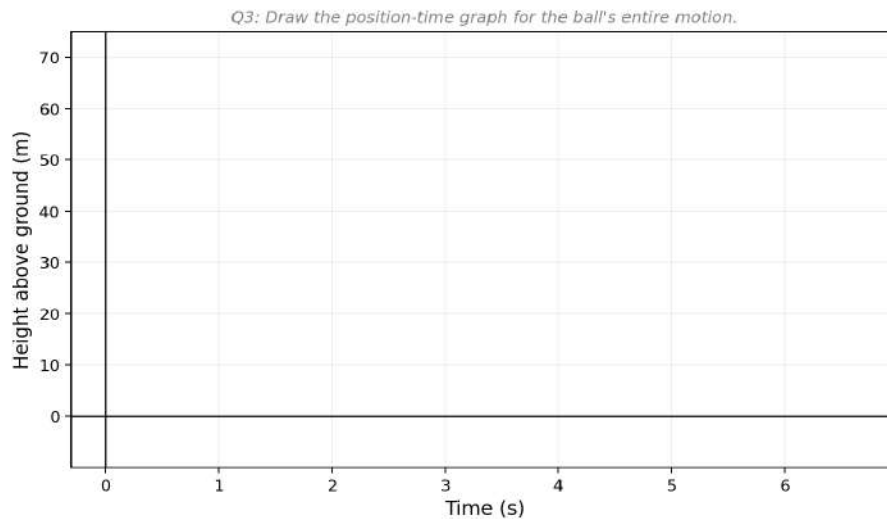
- 2.1 Draw a labelled free body diagram for Block A (showing all forces). (4)
- 2.2 Calculate the tension T in the string. (5)
- 2.3 Calculate the acceleration of the system. (3)
- 2.4 State Newton's Third Law of Motion in words. (2)
- 2.5 Give the Newton's Third Law force pair to the tension T on Block A. (2)
- 2.6 A lubricant is applied to the surface (friction coefficient $\rightarrow 0$). How does the acceleration of the system change? Explain. (1)

[17]

QUESTION 3

Vertical Projectile Motion

A ball is thrown vertically upward from the edge of a cliff 45 m high with an initial velocity of $20 \text{ m}\cdot\text{s}^{-1}$. The ball eventually lands at the base of the cliff. Take upward as positive. $g = 10 \text{ m}\cdot\text{s}^{-2}$.



- 3.1** Calculate the maximum height of the ball above the ground. (3)
- 3.2** Calculate the time taken to reach maximum height. (2)
- 3.3** Calculate the total time of flight until the ball hits the ground. (3)
- 3.4** Calculate the velocity of the ball when it hits the ground. (2)
- 3.5** Draw the position-time graph (y vs t) for the entire flight. Use the axes provided. Label key values clearly. (4)

[14]

QUESTION 4

Momentum and Impulse

A 0.15 kg cricket ball travelling at $30 \text{ m}\cdot\text{s}^{-1}$ horizontally is hit by a bat. The ball leaves at $40 \text{ m}\cdot\text{s}^{-1}$ in the opposite direction.

4.1 Calculate the change in momentum of the ball. (3)

4.2 The bat is in contact with the ball for 0.005 s. Calculate the average force exerted by the bat on the ball. (3)

4.3 State the direction of this force. (1)

A 4 kg trolley A moving east at $6 \text{ m}\cdot\text{s}^{-1}$ collides with a stationary 2 kg trolley B. After collision they move together.

4.4 Calculate the velocity of the combined trolleys after the collision. (3)

4.5 Calculate the kinetic energy lost in this collision. (3)

[13]

QUESTION 5

Work, Energy and Power

A 2 000 kg car engine delivers a driving force of 5 000 N. The car experiences a constant frictional force of 1 200 N. Take $g = 10 \text{ m}\cdot\text{s}^{-2}$.

5.1 Calculate the net force on the car. (1)

5.2 Calculate the acceleration of the car. (2)

5.3 The car accelerates from rest. Calculate the speed after travelling 400 m. (3)

5.4 Calculate the work done by the engine over this 400 m. (2)

5.5 Calculate the work done against friction over 400 m. (1)

5.6 Verify using the work-energy theorem that your answer in 5.3 is correct. (3)

[12]

QUESTION 6

Doppler Effect

A train horn emits a sound of frequency 850 Hz. The speed of sound is $340 \text{ m}\cdot\text{s}^{-1}$. The train travels at $25 \text{ m}\cdot\text{s}^{-1}$ toward a station.

- 6.1** Calculate the frequency heard by a person standing on the platform as the train approaches. (3)
- 6.2** Calculate the frequency heard by the person as the train moves away at the same speed. (3)
- 6.3** How does the wavelength of the sound in FRONT of the approaching train compare to the wavelength of the emitted sound? Explain. (2)
- 6.4** Give ONE application of the Doppler effect used by traffic authorities. (2)
- 6.5** A learner says: "The Doppler shift occurs because the speed of sound changes." Is this correct? Explain. (1)

[11]

QUESTION 7

Electrostatics

Two point charges: $Q_1 = +5 \times 10^{-6} \text{ C}$ and $Q_2 = +5 \times 10^{-6} \text{ C}$ are placed 0.4 m apart. $k = 9 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$.

- 7.1** Calculate the electrostatic force between Q_1 and Q_2 . State whether attractive or repulsive. (4)
- 7.2** A third charge $Q_3 = -2 \times 10^{-6} \text{ C}$ is placed midway between Q_1 and Q_2 . Calculate the net force on Q_3 . State the direction. (5)
- 7.3** Calculate the electric field strength at the midpoint between Q_1 and Q_2 due to Q_1 only. (3)
- 7.4** Draw and describe the electric field pattern around Q_1 alone. (3)

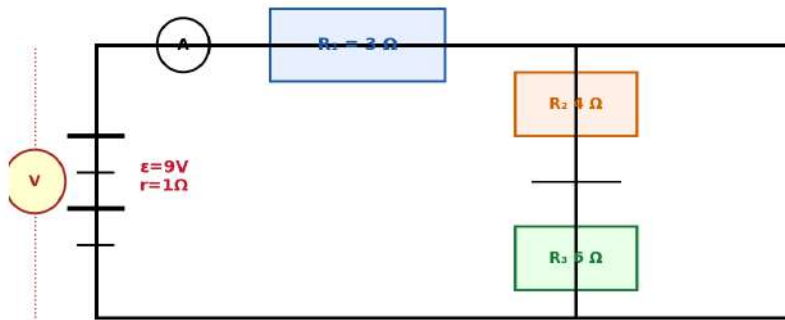
[15]

QUESTION 8

Electric Circuits

A battery with emf 9 V and internal resistance $1\ \Omega$ is connected to an external circuit. $R_1 = 3\ \Omega$ is in series with the parallel combination of $R_2 = 4\ \Omega$ and $R_3 = 6\ \Omega$. A voltmeter is connected across the battery. Refer to the circuit diagram.

Paper B – Q8: Electric Circuit (MEMO)



- 8.1 Calculate the equivalent resistance of R_2 and R_3 in parallel. (2)
- 8.2 Calculate the total external resistance. (2)
- 8.3 Calculate the current from the battery. (3)
- 8.4 Calculate the reading on the voltmeter. (3)
- 8.5 Calculate the potential difference across the parallel combination. (2)
- 8.6 Calculate the current through R_3 . (3)
- 8.7 Calculate the power delivered by the battery. (2)
- 8.8 If R_1 is increased, what happens to the terminal voltage? Explain. (3)

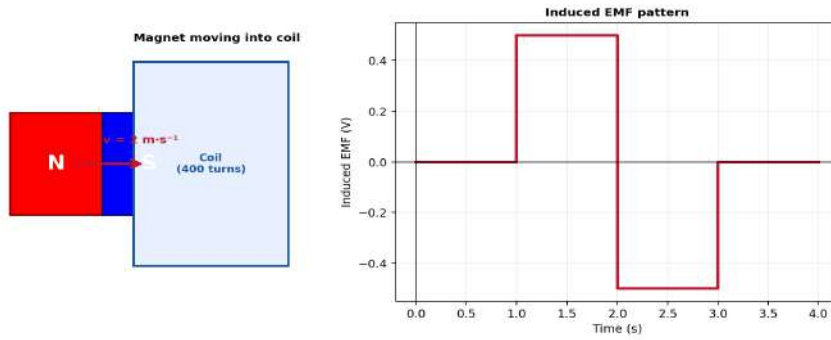
[20]

QUESTION 9

Electromagnetic Induction

A rectangular coil of 400 turns has an area of 0.05 m^2 . It rotates in a uniform magnetic field of 0.8 T at 50 revolutions per second.

- 9.1 State Lenz's Law. (2)
- 9.2 Calculate the peak (maximum) induced EMF. (3)
- 9.3 Calculate the rms EMF. (2)
- 9.4 A bar magnet is thrust quickly into a coil connected to a galvanometer. The galvanometer deflects. Explain why, using Faraday's Law. (3)



9.5 The magnet is then held stationary inside the coil. Explain what happens to the galvanometer reading and why. (2)

9.6 State ONE way to increase the induced EMF in the coil. (2)

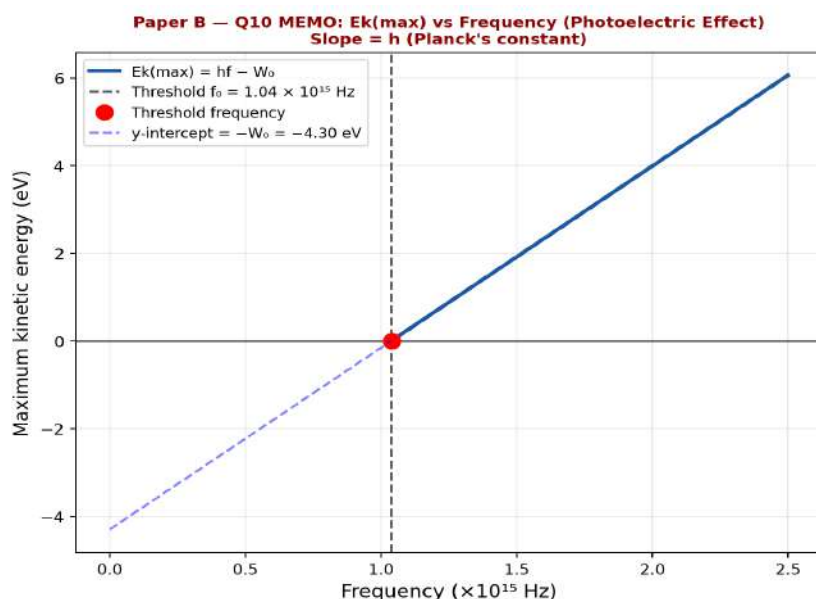
9.7 Compare a step-up transformer to a step-down transformer. Give ONE difference. (1)

[15]

QUESTION 10

Photoelectric Effect

The graph below shows maximum kinetic energy (E_k) of emitted photoelectrons vs the frequency of incident light for a metal surface. The threshold frequency is read as 1.04×10^{15} Hz. Planck's constant $h = 6.63 \times 10^{-34}$ J·s.



10.1 Explain what the threshold frequency represents in terms of the photoelectric effect. (2)

10.2 Read from the graph and calculate the work function W_0 of this metal. (3)

10.3 From the graph, determine the value of Planck's constant. Show your method clearly (use the gradient of the graph). (4)

10.4 Light of wavelength 200 nm is shone on the metal surface. (Speed of light $c = 3 \times 10^8$ m·s $^{-1}$). Calculate the maximum kinetic energy of the emitted photoelectrons. (4)

[13]

TOTAL: 150 marks

PREDICTIVE 2026 PHYSICAL SCIENCES PAPER 1 (PHYSICS)

PAPER B — MEMORANDUM

Accept alternative correct methods. Award consequential marks where applicable.

QUESTION 1 [20 marks]

- | | | |
|----------------|---|-----|
| 1.1 | B — The gravitational force the book exerts on Earth (Newton's 3rd law: equal & opposite) | (2) |
| 1.2 | B — $mg \sin\theta$ (component of gravity along the frictionless slope = net force) | (2) |
| 1.3 | C — acceleration ($g = 10 \text{ m}\cdot\text{s}^{-2}$ downward throughout; speed, velocity, EK all change) | (2) |
| 1.4 | B — Total momentum is conserved if no external net force acts | (2) |
| 1.5 | C — $3 \text{ m}\cdot\text{s}^{-2}$ ($v^2 = u^2 + 2as$: $0 = 900 - 2a \times 150$; $a = 900/300 = 3 \text{ m}\cdot\text{s}^{-2}$) | (2) |
| 1.6 | B — lower than emitted (source moving away → Doppler effect → lower frequency) | (2) |
| 1.7 | A — kQ/r^2 (definition of electric field strength $E = F/q = kQ/r^2$) | (2) |
| 1.8 | B — no current flows ($V_{\text{terminal}} = \mathcal{E} - Ir = \mathcal{E}$ when $I = 0$) | (2) |
| 1.9 | C — Photoelectric effect (demonstrates particle/photon nature of light) | (2) |
| 1.10 | C — hf_0 ($E_k = hf - W_0 = h(2f_0) - hf_0 = hf_0$) | (2) |
| [TOTAL Q1: 20] | | |

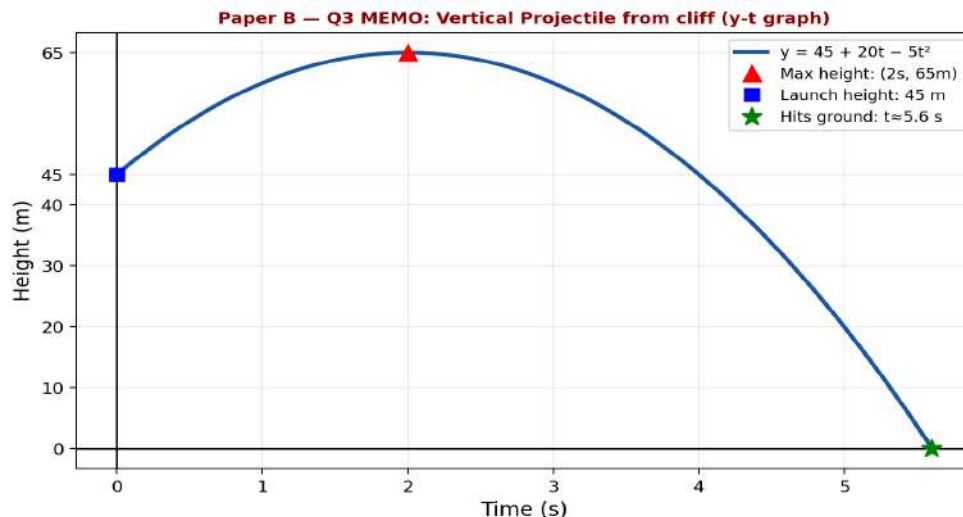
QUESTION 2 [17 marks]

- | | | |
|-----|---|-----|
| 2.1 | FBD for Block A must show:
Weight $W_A = m_A g = 80 \text{ N}$ downward
Normal force $N = 80 \text{ N}$ upward (perpendicular to surface)
Tension T rightward (toward pulley)
Friction $f_k = \mu_k \times N$ leftward (opposing motion) | (4) |
| 2.2 | Apply Newton's 2nd law to system (A + B, same acceleration a):
Net force = $m_B \times g - f_k = m_B \times g - \mu_k \times m_A \times g$
$= 3 \times 10 - 0.25 \times 8 \times 10 = 30 - 20 = 10 \text{ N}$
Total mass = $8 + 3 = 11 \text{ kg}$
$a = 10/11 = 0.909 \text{ m}\cdot\text{s}^{-2}$
For Block B: $m_B \times g - T = m_B \times a$
$T = m_B(g - a) = 3(10 - 0.909) = 3 \times 9.091$
$T = 27.27 \text{ N}$ | (5) |
| 2.3 | $a = F_{\text{net}} / m_{\text{total}} = 10/11$
$a = 0.91 \text{ m}\cdot\text{s}^{-2}$ (downward for B, along surface for A) | (3) |
| 2.4 | Newton's Third Law: If object A exerts a force on object B, then object B exerts an equal in magnitude but opposite in direction force on object A. (Forces always occur in pairs.) | (2) |

- 2.5 The reaction force to T (tension pulling A toward pulley) is: (2)
T on Block A is the string pulling A → reaction is A pulling
string back (toward left).
OR: T acts on A (rightward); Newton's 3rd pair = A exerts
force on string (leftward)
- 2.6 $f_k \rightarrow 0$, so net force = $m_B \times g = 30 \text{ N}$ (1)
 $a = 30/11 = 2.73 \text{ m}\cdot\text{s}^{-2}$ — acceleration INCREASES

[TOTAL Q2: 17]

QUESTION 3 [14 marks]



- 3.1 At max height, $v = 0$; $y_0 = 45$ m (cliff height) (3)
 $v^2 = v_0^2 + 2a\Delta y$: $0 = (20)^2 + 2(-10)(\Delta y)$
 $\Delta y = 400/20 = 20$ m above cliff
 Max height above ground = $45 + 20 = 65$ m
- 3.2 $v = v_0 + at$: $0 = 20 - 10t$ (2)
 $t = 2.0$ s (to reach max height)
- 3.3 $y = y_0 + v_0t - \frac{1}{2}gt^2$; when $y = 0$ (ground): (3)
 $0 = 45 + 20t - 5t^2 \rightarrow 5t^2 - 20t - 45 = 0 \rightarrow t^2 - 4t - 9 = 0$
 $t = (4 + \sqrt{(16 + 36)})/2 = (4 + \sqrt{52})/2 = (4 + 7.211)/2$
 $t = 5.61$ s (total time of flight)
- 3.4 $v^2 = v_0^2 + 2a\Delta y$: $v^2 = (20)^2 + 2(-10)(0 - 45) = 400 + 900 = 1300$ (2)
 $v = -\sqrt{1300} = -36.06$ m·s⁻¹
 Speed on impact = 36.06 m·s⁻¹ DOWNWARD
- 3.5 y-t graph: parabola starting at (0; 45), rising to (2; 65), then (4)
 curving down to (5.61; 0)
 ✓ correct starting height 45 m ✓ max at (2; 65) ✓ ends at ground ✓ curve shape
- [TOTAL Q3: 14]

QUESTION 4 [13 marks]

- 4.1 Take direction ball travels initially as positive (+30 m·s⁻¹) (3)
 $\Delta p = m(v_f - v_i) = 0.15(-40 - 30) = 0.15 \times (-70)$
 $\Delta p = -10.5$ kg·m·s⁻¹ (magnitude = 10.5 kg·m·s⁻¹)
- 4.2 $F = \Delta p / \Delta t = 10.5 / 0.005$ (3)
 $F = 2100$ N
- 4.3 Force is in the direction opposite to the initial velocity of the (1)
 ball (i.e., in the direction the ball travels after being hit).
- 4.4 Conservation of momentum: $m_A \times v_A = (m_A + m_B)v_f$ (3)
 $4 \times 6 = (4 + 2)v_f$
 $24 = 6v_f$
 $v_f = 4$ m·s⁻¹ (eastward)

4.5

$$EK_i = \frac{1}{2}(4)(6^2) = \frac{1}{2} \times 4 \times 36 = 72 \text{ J}$$

$$EK_f = \frac{1}{2}(6)(4^2) = \frac{1}{2} \times 6 \times 16 = 48 \text{ J}$$

$$\Delta EK_{\text{lost}} = 72 - 48 = 24 \text{ J}$$

(3)

[TOTAL Q4: 13]

QUESTION 5 [12 marks]

- 5.1 $F_{\text{net}} = F_{\text{drive}} - F_{\text{friction}} = 5000 - 1200$ (1)
 $F_{\text{net}} = 3800 \text{ N (forward)}$
- 5.2 $a = F_{\text{net}}/m = 3800/2000$ (2)
 $a = 1.9 \text{ m}\cdot\text{s}^{-2}$
- 5.3 $v^2 = u^2 + 2as = 0 + 2(1.9)(400) = 1520$ (3)
 $v = \sqrt{1520}$
 $v = 38.99 \text{ m}\cdot\text{s}^{-1} \approx 39 \text{ m}\cdot\text{s}^{-1}$
- 5.4 $W_{\text{engine}} = F_{\text{drive}} \times d = 5000 \times 400$ (2)
 $W_{\text{engine}} = 2\,000\,000 \text{ J} = 2 \text{ MJ}$
- 5.5 $W_{\text{friction}} = f \times d = 1200 \times 400$ (1)
 $W_{\text{friction}} = 480\,000 \text{ J} = 480 \text{ kJ}$
- 5.6 $W_{\text{net}} = W_{\text{engine}} - W_{\text{friction}} = 2\,000\,000 - 480\,000 =$ (3)
 $1\,520\,000 \text{ J}$
 $\Delta E_K = \frac{1}{2}mv^2 - 0 = \frac{1}{2} \times 2000 \times 1520 = 1\,520\,000 \text{ J}$
 $W_{\text{net}} = \Delta E_K \checkmark \text{ (work-energy theorem verified)}$

[TOTAL Q5: 12]

QUESTION 6 [11 marks]

- 6.1 $f_L = v/(v - v_s) \times f_s = 340/(340 - 25) \times 850$ (3)
 $= 340/315 \times 850$
 $f_L = 917.46 \text{ Hz}$
- 6.2 $f_L = v/(v + v_s) \times f_s = 340/(340 + 25) \times 850$ (3)
 $= 340/365 \times 850$
 $f_L = 791.78 \text{ Hz}$
- 6.3 Wavelength is SHORTER in front of the approaching train. (2)
The train 'catches up' to the waves it emits, compressing them,
so wavelength $= (v - v_s)/f < v/f$ (shorter than emitted wavelength).
- 6.4 Speed traps / radar guns — traffic police use Doppler radar to (2)
measure the speed of vehicles.
- 6.5 No, this is incorrect. The speed of sound in a given medium (1)
remains constant. It is the wavelength (and hence frequency)
that changes, NOT the speed of the wave.

[TOTAL Q6: 11]

QUESTION 7 [15 marks]

- 7.1 $F = kQ_1Q_2/r^2 = (9 \times 10^9 \times 5 \times 10^{-6} \times 5 \times 10^{-6}) / (0.4)^2$ (4)
 $= (9 \times 10^9 \times 25 \times 10^{-12}) / 0.16$
 $= 225 \times 10^{-3} / 0.16$
 $F = 1.406 \text{ N} - \text{REPULSIVE (both charges positive)}$
- 7.2 Q_3 is 0.2 m from each charge (midpoint). (5)
 $F_1 \text{ on } Q_3 \text{ from } Q_1 = k|Q_1||Q_3|/r^2 = 9 \times 10^9 \times 5 \times 10^{-6} \times 2 \times 10^{-6} / (0.2)^2$
 $= 90 \times 10^{-3} / 0.04 = 2.25 \text{ N (attractive toward } Q_1)$
 $F_2 \text{ on } Q_3 \text{ from } Q_2 = \text{same calculation} = 2.25 \text{ N (attractive toward } Q_2)$
Both forces act in opposite directions (Q_3 is pulled equally to left and right).
Net force on $Q_3 = 2.25 - 2.25 = 0 \text{ N (by symmetry)}$
- 7.3 $E = kQ_1/r^2 = 9 \times 10^9 \times 5 \times 10^{-6} / (0.2)^2$ (3)
 $= 45000 / 0.04$
 $E = 1.125 \times 10^6 \text{ N} \cdot \text{C}^{-1}$ directed away from Q_1 (toward Q_2)
- 7.4 Electric field around Q_1 (positive charge): field lines radiate outward in all directions. (3)
Lines point away from the positive charge.
Field is stronger closer to Q_1 (lines denser near charge).
- [TOTAL Q7: 15]

QUESTION 8 [20 marks]

- 8.1 $1/R_p = 1/R_2 + 1/R_3 = 1/4 + 1/6 = 3/12 + 2/12 = 5/12$ (2)
 $R_p = 12/5 = 2.4 \Omega$
- 8.2 $R_{\text{ext}} = R_1 + R_p = 3 + 2.4 = 5.4 \Omega$ (2)
- 8.3 $I = \epsilon / (R_{\text{ext}} + r) = 9 / (5.4 + 1) = 9 / 6.4$ (3)
 $I = 1.406 \text{ A}$
- 8.4 Voltmeter reads terminal voltage $= \epsilon - Ir = 9 - 1.406 \times 1$ (3)
 $V_{\text{terminal}} = 7.594 \text{ V} \approx 7.59 \text{ V}$
- 8.5 $V_{\text{parallel}} = I \times R_p = 1.406 \times 2.4$ (2)
 $V_{\text{parallel}} = 3.375 \text{ V} \approx 3.38 \text{ V}$
- 8.6 $I_3 = V_{\text{parallel}} / R_3 = 3.375 / 6$ (3)
 $I_3 = 0.5625 \text{ A} \approx 0.56 \text{ A}$
- 8.7 $P = \epsilon I = 9 \times 1.406$ (2)
 $P = 12.65 \text{ W}$
- 8.8 If R_1 increases $\rightarrow R_{\text{total}}$ increases $\rightarrow$ current I decreases $\rightarrow$ (3)
'lost voltage' across $r (= Ir)$ decreases
 $\therefore$ Terminal voltage $= \epsilon - Ir$ INCREASES
So with higher R_1 , terminal voltage increases (closer to $\epsilon = 9 \text{ V}$)
- [TOTAL Q8: 20]

QUESTION 9 [15 marks]

- 9.1 Lenz's Law: The induced current flows in a direction such that the magnetic field it creates opposes the change in magnetic flux that caused it. (2)
- 9.2 Peak EMF = $NBA\omega$ (3)
 $\omega = 2\pi \times 50 = 100\pi \text{ rad}\cdot\text{s}^{-1}$
 $E_{\text{peak}} = 400 \times 0.8 \times 0.05 \times 100\pi$
 $= 400 \times 0.8 \times 0.05 \times 314.16$
 $E_{\text{peak}} = 5026.5 \text{ V} \approx 5027 \text{ V}$
- 9.3 $E_{\text{rms}} = E_{\text{peak}}/\sqrt{2} = 5026.5/1.4142$ (2)
 $E_{\text{rms}} = 3554 \text{ V}$
- 9.4 As the magnet moves into the coil, the magnetic flux through the coil increases. (3)
By Faraday's Law, this changing flux induces an EMF in the coil.
This EMF drives a current through the circuit (coil + galvanometer), causing deflection.
- 9.5 When the magnet is stationary, the magnetic flux is not changing ($d\Phi/dt = 0$). (2)
By Faraday's Law, no EMF is induced $\rightarrow$ no current flows $\rightarrow$ galvanometer reads zero.
- 9.6 ANY ONE: Increase N (more turns) / increase B (stronger magnet) / increase A (larger coil) / increase ω (rotate faster). (2)
- 9.7 Step-up: output voltage $>$ input voltage ($N_s > N_p$); output current $<$ input current. (1)
Step-down: output voltage $<$ input voltage ($N_s < N_p$); output current $>$ input current.

[TOTAL Q9: 15]

QUESTION 10 [13 marks]

- 10.1 The threshold frequency is the minimum frequency of incident light required to emit electrons from the metal surface. Below this frequency, no photoelectrons are emitted regardless of intensity. It corresponds to the frequency at which $hf = W_0$. (2)
- 10.2 From graph: at threshold frequency $f_0 = 1.04 \times 10^{15} \text{ Hz}$, $E_k = 0$ (3)
 $W_0 = h \times f_0 = 6.63 \times 10^{-34} \times 1.04 \times 10^{15}$
 $W_0 = 6.895 \times 10^{-19} \text{ J} \approx 6.90 \times 10^{-19} \text{ J}$
- 10.3 Gradient of E_k vs f graph = h (Planck's constant) (4)
Using two points from the graph (e.g. at $f = 1.5 \times 10^{15} \text{ Hz}$, $E_k \approx 3.07 \times 10^{-19} \text{ J}$ and $f_0 = 1.04 \times 10^{15}$
Gradient = $\Delta E_k / \Delta f = (3.07 \times 10^{-19} - 0) / (1.5 \times 10^{15} - 1.04 \times 10^{15})$
 $= 3.07 \times 10^{-19} / 0.46 \times 10^{15}$
 $= 6.67 \times 10^{-34} \text{ J}\cdot\text{s} \approx 6.63 \times 10^{-34} \text{ J}\cdot\text{s} \checkmark$ (consistent with Planck's constant)
- 10.4 $f = c/\lambda = (3 \times 10^8) / (200 \times 10^{-9}) = 3 \times 10^8 / 2 \times 10^{-7} = 1.5 \times 10^{15} \text{ Hz}$ (4)
 $E_{\text{photon}} = hf = 6.63 \times 10^{-34} \times 1.5 \times 10^{15} = 9.945 \times 10^{-19} \text{ J}$
 $E_k(\text{max}) = E_{\text{photon}} - W_0 = 9.945 \times 10^{-19} - 6.895 \times 10^{-19}$
 $E_k(\text{max}) = 3.05 \times 10^{-19} \text{ J}$

[TOTAL Q10: 13]

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	TOTAL
20	17	14	13	12	11	15	20	15	13	150